HyMAGE: Semantic-Aware Dynamic Hypergraph Generation

Bingqiao Gu[♠] Jiale Zeng[♠] Nuoran Zhou[♢] Xingqin Qi[♠] Dong Li[♠,*]

[♠]School of Mathematics and School of Airspace Science and Engineering, Shandong University [♢]Australian National University

{gubingqiao, zengjiale}@mail.sdu.edu.cn u8159865@anu.edu.au {qixingqin, dongli}@sdu.edu.cn

Code & data: https://doi.org/10.5281/zenodo.20496905

Motivation

Complex networks traditionally capture pairwise links, yet many real systems are driven by *high-order* interactions involving more than two entities—structures known as *hypernetworks*. In online social networks, information propagates through group chats; in collaboration networks, papers are co-authored by several researchers; in biological systems, protein complexes act as one unit. **Hypergraphs** allow a single edge to connect an arbitrary number of nodes and have become the principal mathematical tool for modeling such multi-way interactions.

A fundamental question is how these structures form and evolve. Hypernetwork evolution models serve two critical roles. (1) As **theoretical tools**, they uncover the mechanisms of high-order structural evolution and explain social and natural phenomena. (2) As **data generation tools**, they produce large-scale synthetic hypergraphs whose statistics match real networks—needed for training hypergraph learners, privacy-preserving digital twins, knowledge distillation into graph structure, and scalable environments for diffusion tasks such as influence maximization.

Existing generators span configuration and stochastic block models, preferential-attachment evolution, and algorithms targeting overlap or temporal dynamics. Despite this progress they share two key limitations:

1. **Attribute-blind.** Dynamics are inferred from topology alone, ignoring the node attributes — research interests, chemical mechanisms — that actually drive who joins whom.
2. **Passive nodes.** Nodes execute fixed, immutable rules, whereas real entities are autonomous agents endowed with perception, decision-making and adaptation. Macroscopic network properties emerge bottom-up from such microscopic cooperation.

To close this gap we use LLMs inside a multi-agent system, motivated by two observations. (1) LLM text generation and knowledge reserves can supply rich, domain-specific node attributes, while their semantic understanding acts as a *dynamic semantic model*: hyperedges form and dissolve from local semantic and structural context rather than fixed centralized rules. (2) Multi-agent systems reconceptualize nodes as autonomous agents that adjust strategies from the local environment, reproducing the emergent phenomena of real complex systems. PA is only a *weak structural prior*; the LLM is a *local* selection module, not a global generator or a training objective. Semantic-driven selection and structure-driven connection bias act in concert, so hyperedges form or dissolve according to node states and local context.

HyMAGE (**Hy**pergraph generation via **M**ulti-**A**gent **G**enerative **E**volution) casts every node as an autonomous agent and queries an LLM as a *local semantic oracle*, so that hyperedges form and dissolve in a self-organizing way — with *no* graph-structure pretraining and *no* centralized objective.

The HyMAGE Framework

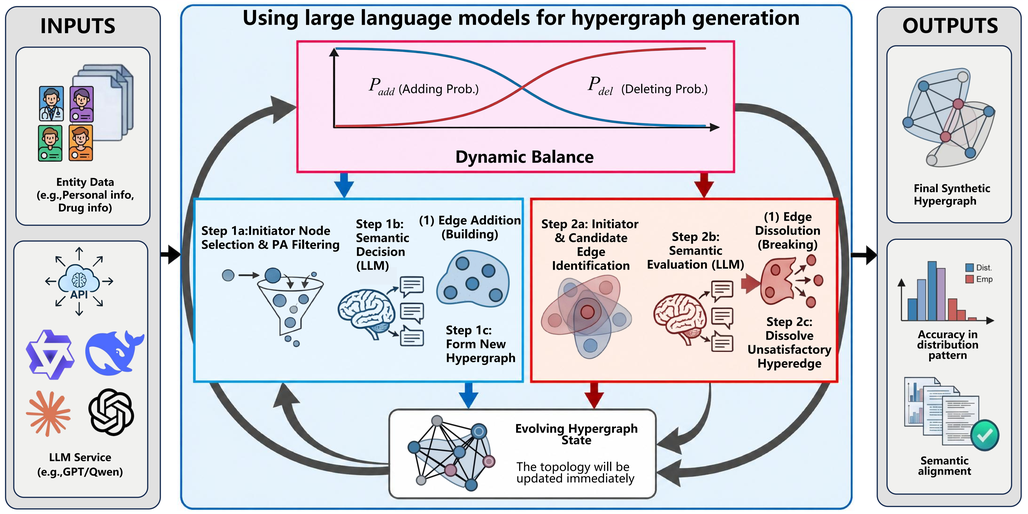

Figure 1: Core mechanism of HyMAGE. Given entity attributes and an LLM service, HyMAGE iteratively balances hyperedge *addition* and *dissolution* to produce a semantically aligned synthetic hypergraph.

Two node-initiated operators drive the evolution:

- **Formation.** A node recruits collaborators from a preferential-attachment candidate pool (structural *visibility*); the LLM *selects* the semantically compatible ones.
- **Dissolution.** A node re-evaluates its incident hyperedges by *coherence*, *value* and *conflict*, and drops those that no longer serve it.

Nine Universal Patterns

Real hypergraphs exhibit nine universal patterns that we use as fidelity benchmarks. *Static snapshot patterns* (P1–P6) are measured from a single snapshot; *dynamic evolutionary patterns* (P7–P9) require time-ordered hyperedges. This taxonomy separates what a generator must reproduce in a final snapshot from how the hypergraph should evolve.

Dimension	Patterns	Measurement focus
Static	P1–P6	Snapshot topology: node degree, hypercoreness, hyperedge size, pairwise hyperedge intersection, incidence spectrum, and group degree.
Dynamic	P7–P9	Evolutionary behavior: temporal locality, bursty persistence, and overlap-density decay.

Table 1: Taxonomy of the nine universal hypergraph patterns.

- **P1. Heavy-tailed degree.** A few hubs join many hyperedges; most nodes stay sparse — the "rich-get-richer" mechanism of real hypernetworks.
- **P2. Heavy-tailed hypercoreness.** HC(v) measures depth in nested dense cores; its distribution is likewise heavy-tailed.
- **P3. Heavy-tailed hyperedge size.** Most groups are small; a few interactions are very large.
- **P4. Heavy-tailed intersection size.** Most hyperedge pairs share little; a few exhibit large overlaps (strong structural correlation).
- **P5. Skewed singular values.** Incidence-matrix singular values are heavily skewed, implying an approximately low-rank structure.
- **P6. Heavy-tailed group degree.** A few communities are repeatedly active; many remain small, tightly knit groups.
- **P7. Temporal locality.** Recently active nodes reappear in subsequent hyperedges — short-term memory in the evolutionary process.
- **P8. Power-law persistence.** Inter-event times follow a power law: long inactivity punctuated by bursty activity.
- **P9. Diminishing overlaps.** New hyperedges overlap existing ones less and less. Density of interactions $DoI(G_t) = |\{\{e_i, e_j\} \mid e_i \cap e_j \neq \emptyset\}|/|\{\{e_i, e_j\}\}|$ decays over time.

Semantic Preferential Attachment

With progress $p_t = m_t/M$, sigmoid rates let formation dominate early and dissolution take over as the hypergraph matures:

$$\lambda_{\text{form}}(p) = \alpha_f\,\sigma\big(-k_f(p-\theta_f)\big) + \beta_f, \quad \lambda_{\text{diss}}(p) = \alpha_d\,\sigma\big(k_d(p-\theta_d)\big) + \beta_d. \tag{1}$$

Initiator by PA, collaborators by the LLM over pool $\mathcal{C}$ — *structure proposes, semantics disposes*:

$$P(v_c = i) = \frac{d_i(t)+\alpha}{\sum_{j\in V}(d_j(t)+\alpha)}, \quad P(j \in e_{\text{new}} \mid v_c, \mathcal{C}) = \pi_{\text{LLM}}(j \mid \mathbf{x}_{v_c}, \{(\mathbf{x}_u, d_u)\}_{u\in\mathcal{C}}). \tag{2}$$

A hyperedge is dropped with $P_{\text{diss}}(e \mid v_c) = (1 - C(e))\lambda_{\text{diss}}(p_t)$, where $C(e)$ is mean pairwise semantic coherence. Mean-field reduction yields fitness-based PA, $d_i \propto (r_i + c)^{-\gamma}$, so heavy tails emerge without being fitted.

Structural Fidelity

Method	Venue	Deg.	HCore.	Int.	SV	IP	Grp.	Pers.	Avg.
HyperCL	KDD'20	0.122	0.139	0.043	0.026	0.013	0.119	0.102	0.079
HyperFF	ICDM'20	0.238	0.224	0.046	0.049	0.024	0.082	0.067	0.104
HyperPA	KDD'20	0.342	0.345	0.059	0.041	0.010	0.067	0.108	0.139
HyperLAP	WWW'21	0.132	0.146	0.022	0.034	0.014	0.052	0.054	0.065
HyRec	WWW'25	0.732	0.845	0.336	0.028	0.011	0.089	0.416	0.351
HyperPLR	ICLR'25	0.574	0.697	0.039	0.087	0.013	0.139	0.140	0.241
HyMAGE (Ours)	–	**0.105**	**0.127**	0.028	0.029	**0.009**	**0.049**	**0.048**	**0.056**

Table 2: Averaged over **14 real hypergraphs**: JS divergence for distributional patterns, $1 - r$ for trends, **lower is better** (best, second, third).

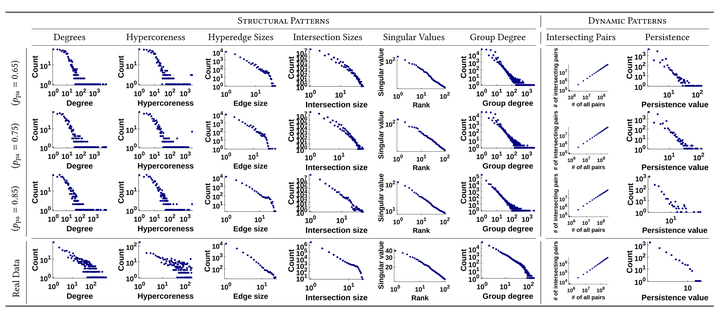

Figure 2: Real hypergraph vs. HyMAGE under three attachment probabilities (email-Eu, full scale). All eight measurable patterns are reproduced, most faithfully at $p_{\text{pa}} = 0.85$.

Downstream Utility

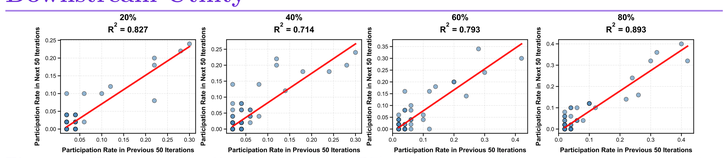

Figure 3: Temporal locality verification (P7). Historical vs. future node participation at four checkpoints; average $R^2 = 0.81$.

Diffusion simulation. We test whether generated hypergraphs support downstream diffusion analysis via influence maximization on coauth-CS under the Hypergraph Independent Cascade (HIC) model, where influence spreads through crowd-level activation inside hyperedges. Real and synthetic spread curves are compared under four seeding strategies (Degree, PageRank, Closeness, Betweenness), with fidelity measured by NRMSE. Across all four strategies the HyMAGE hypergraph (bold colored) most closely follows the real-data curve (bold black) and attains the lowest NRMSE, showing that HyMAGE preserves diffusion-relevant structure beyond the static pattern metrics.

- **HGNN training data.** Hypergraph neural networks trained *solely* on HyMAGE data transfer to real hypergraphs, surpassing both real training topology and every baseline generator.

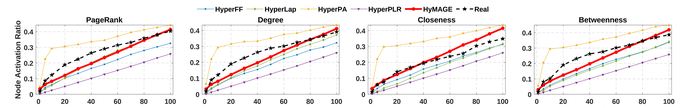

Figure 4: Influence maximization on coauth-CS (HIC model), one panel per seeding strategy. HyMAGE tracks the real hypergraph (bold black) most closely, with the lowest NRMSE.

- **Ablations.** The full model beats the *Embedding Oracle* (Sentence-BERT similarity) and *w/o LLM* (pure PA) across four backbone LLMs.

Conclusion

We introduced HyMAGE, a semantic-aware hypergraph generation framework that reframes hypernetwork evolution as multi-agent generative evolution. By casting nodes as autonomous agents and employing an LLM as a local semantic oracle—without graph-structure pretraining or a centralized optimization objective—hyperedge formation and dissolution self-organize from local semantic affinity and structural context. Our key contributions are:

- **Semantic Preferential Attachment.** A principled paradigm that extends classical PA with LLM-approximated semantic scoring and dual formation–dissolution operators, bridging topology-only generation and self-organizing hypergraph evolution.
- **Comprehensive Structural Fidelity.** On 14 datasets spanning social, collaboration and biomedical domains, HyMAGE simultaneously reproduces nine structural patterns (P1–P9) with superior fidelity over all baselines, covering both static snapshot properties and dynamic evolutionary behaviors, and with no user-side training.
- **Downstream Utility.** In a coauth-CS influence maximization case study under the HIC model, HyMAGE-generated hypergraphs produce diffusion curves closest to the real hypergraph among compared generators. HGNNs trained *solely* on synthetic data further achieve high accuracy on real targets, validating their value for data augmentation and algorithm benchmarking.

Benchmark. Together, the nine structural pattern metrics, the reported influence-maximization case study, and the HGNN node-classification protocol constitute—to our knowledge—the first comprehensive benchmark for evaluating synthetic attributed hypergraphs and hypernetwork evolution models.

Future directions. Promising extensions include modeling dynamic node arrival and departure for open-system simulation, generating heterogeneous or signed hypergraphs for richer relational semantics, and fine-tuning domain-specific semantic oracles. HyMAGE also supplies topology-and-semantics-aligned attributed hypergraphs for hypergraph foundation models, whose scaling laws demand diverse multi-domain corpora that are scarce in real data.

Acknowledgements

This work was supported by the National Natural Science Foundation of China under Grant Nos. 62376136, 62076149 and 12471330, and by the Guangdong Basic and Applied Basic Research Foundation under Grant No. 2024A1515011935.